REINFORCEMENT LEARNING · MID-REVIEW

SCoPe: Trust-Region Latent Planning with JEPA World Models

Plan in a learned world model, but act only on imagined futures the model can vouch for.

TEAM

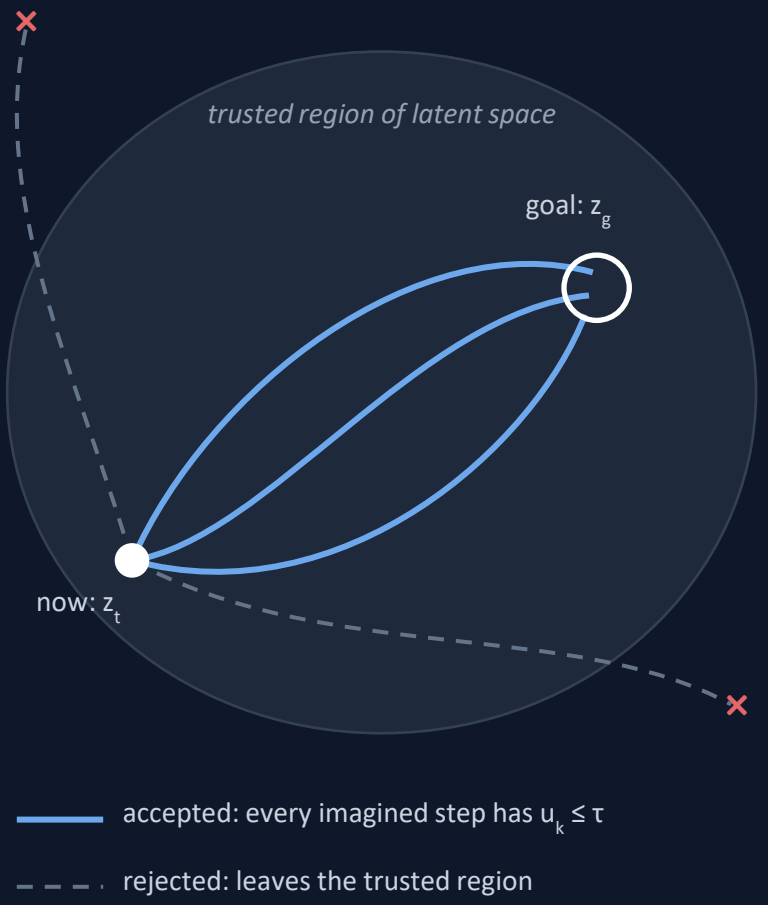
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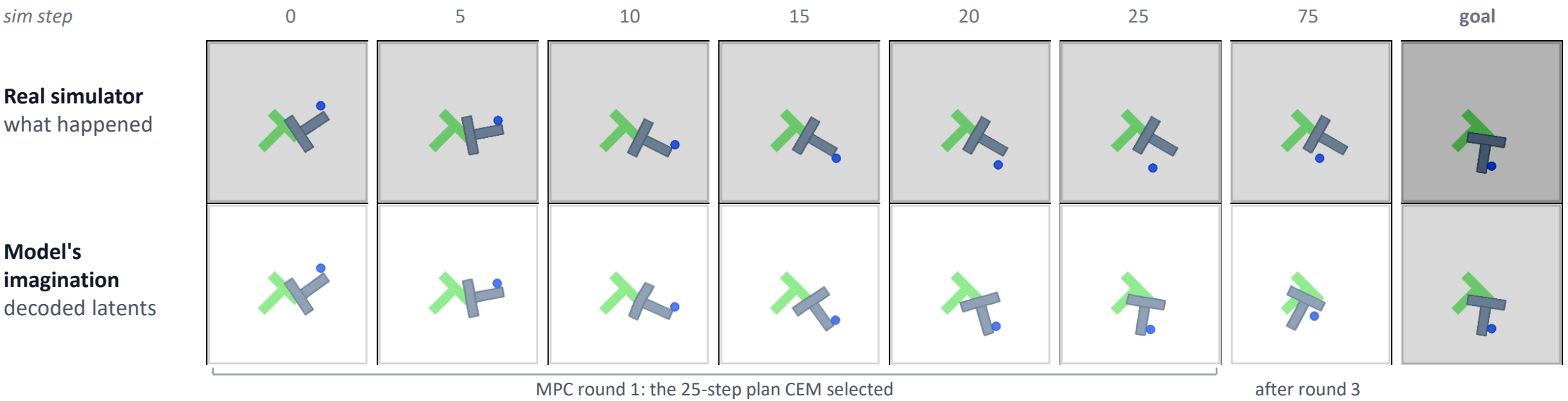
[DEPARTMENT] · [INSTITUTION] · [REVIEW DATE]

RL area model-based RL · planning with a learned latent world model (MPC)



MEASURED PUBLISHED

A world model can imagine a success that never happens



Push-T · seed 101 · episode 0 · our run with the released DINO-WM checkpoint. In imagination the T turns to the goal pose; in the simulator the same actions barely move it.

WHY IT MATTERS

80% vs 15% zero-shot real-arm pick-and-place: V-JEPA 2-AC latent planning vs the Octo policy [2]

70% vs 0% hierarchical vs flat latent planning on real pick-and-place [15]

26% → 34% collisions after adding an uncertainty penalty to MPC [18]

PROBLEM STATEMENT

Latent planners pick the plan whose imagined future looks closest to the goal, with no check that the imagination is reliable; errors compound over the horizon and become real actions. **Proposed:** score every imagined step, reject plans that leave the trusted region, fall back safely, without retraining the world model.

RESEARCH QUESTION

Can an uncertainty score on imagined JEPA latents flag unreliable rollouts well enough that rejecting them improves real planning success, without rejecting most good plans?

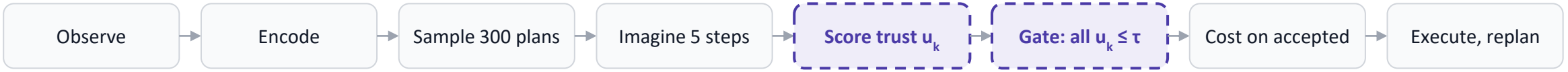
Scope: DINO-WM checkpoints on Push-T, Wall and PointMaze (simulated), frozen encoder, one 8 GB GPU; no real robot.

PUBLISHED PROPOSED

Existing work and our difference: a decision-time trust check

Family and studies	How model error is handled	Reported result	Gap for SCoPe
Uncertainty-aware model-based RL [6, 8–11]	ensembles, short or adaptive rollouts, reward penalties, HALT states	PETS: model-free return with 8× / 125× fewer samples than SAC / PPO	acts in training on low-dimensional states; never vetoes a plan
Latent safety filter [16]	epistemic uncertainty + conformal threshold veto unsafe actions	vision-based Franka arm, simulation and hardware	avoids failure sets with a learned safety value; not goal reaching
Reward-based latent agents [7, 12, 13]	latent world model with CEM, actor-critic or MPPI	DreamerV3: 150+ tasks; TD-MPC2: 104 tasks	need a reward; every imagined rollout is trusted
JEPA latent planners [1, 14]	frozen features + predictor; CEM-MPC to a goal image, reward-free	DINO-WM success 0.90 / 0.96 / 0.98 (Push-T / Wall / PointMaze)	no reliability check. **DINO-WM is our base.**
Real-robot latent planners [2, 15]	action-conditioned predictor; hierarchical MPC	pick-and-place 80% vs 15% (Octo); 70% vs 0% (flat)	target scale and horizon, not reliability
Counter-evidence [17–21]	analyses of uncertainty and latent reliability	disagreement misses compounding error; penalties raised collisions 26% → 34%	a trust score must be tested against real error first (H1)

OUR DIFFERENCE: ONE CHANGE INSIDE THE PLANNER



Dashed violet = new (planned); too few pass → resample → shorter horizon → safe action → stop and flag. DINO-WM today runs the same loop without these two steps. The closest work [16] vetoes actions to avoid failures; SCoPe vetoes goal-reaching plans and first tests the score against real error.

PUBLISHED PROPOSED

RL formulation: a latent-space MDP solved by CEM-based MPC

Element	Our definition (Push-T unless stated)
State / observation	224×224 image + agent position; planning state = frozen DINOv2 [3] patch tokens (196 × 384), 3-frame history
Action	2-D target for the agent (PD-tracked), planned as 10-D chunks of 5 simulator steps; horizon H = 5
Reward / cost	no environment reward: cost = latent distance of the last imagined step to the goal image
Transition	real: 2-D contact physics (pymunk); imagined: ViT predictor f_θ
Episode	one start/goal pair; MPC rounds of 25 simulator steps, at most 3 rounds
Termination	success when ‖Δposition‖ < 20 and |Δθ| < π/9 (Wall < 4.5, PointMaze < 0.5)
Policy, discount	$\pi(o_t, g)$ = first 5 chunks of the best CEM plan, then replan; γ = 1 (terminal cost)

ALGORITHM: CEM-BASED MPC (DINO-WM'S PLANNER)

- 300 sampled plans, refit to the top 30, 30 iterations; execute 5 chunks, replan
- vs PPO / SAC: need a reward and online interaction; our data are offline and reward-free
- vs gradient planning: Push-T 0.28, against 0.86 (one CEM plan) and 0.90 (MPC) [1]
- fits SCoPe: an explicit candidate set to filter; cost ≈ 3.4 min per plan on our GPU

DATA AND SIMULATION (DINO-WM RELEASE [26])

Push-T: pymunk 2-D physics [24], 18.5k offline samples · Wall: 2-D room with a door, 1,920 trajectories · PointMaze: MuJoCo 2.1 [25], 2,000 trajectories

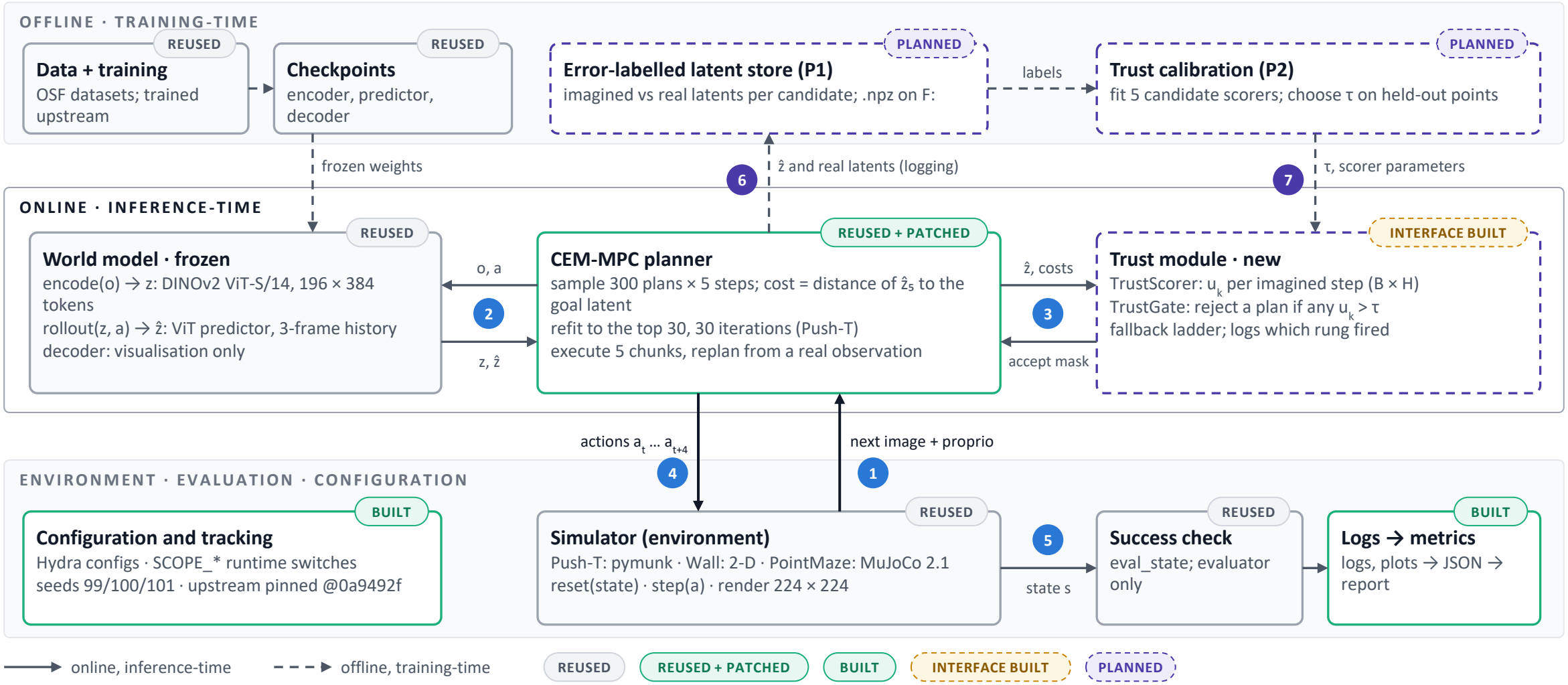
Simulated and reward-free; evaluation pairs come from the validation split or freshly sampled states; seeds 99–101 × 50 pairs.

$$\text{minimise over } a_t \dots a_{t+4}: \quad c_g = \mathrm{MSE}_{\text{visual}}(\hat{z}_{t+5}, z_g) + \alpha \cdot \mathrm{MSE}_{\text{proprio}}(\hat{z}_{t+5}, z_g)$$

$$\text{subject to (SCoPe): } \quad u_k \le \tau \text{ for every imagined step } k = 1 \dots 5; \quad \text{too few candidates pass} \rightarrow \text{fallback ladder}$$

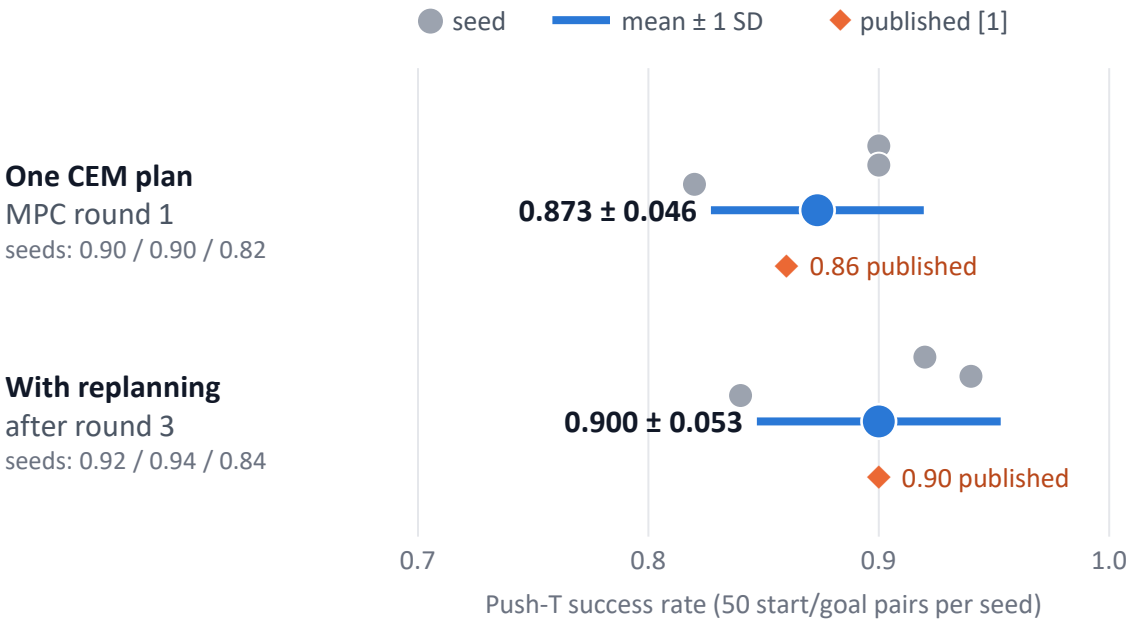
Why RL: a push takes effect steps later and success is judged only after the plan, so actions are chosen as a sequence (model-based RL at decision time). **Reward hacking** here means exploiting model errors, so success is always judged by the simulator. α = 1 (Push-T, Wall) or 0 (PointMaze), upstream values.

System architecture: online agent and offline learning loop



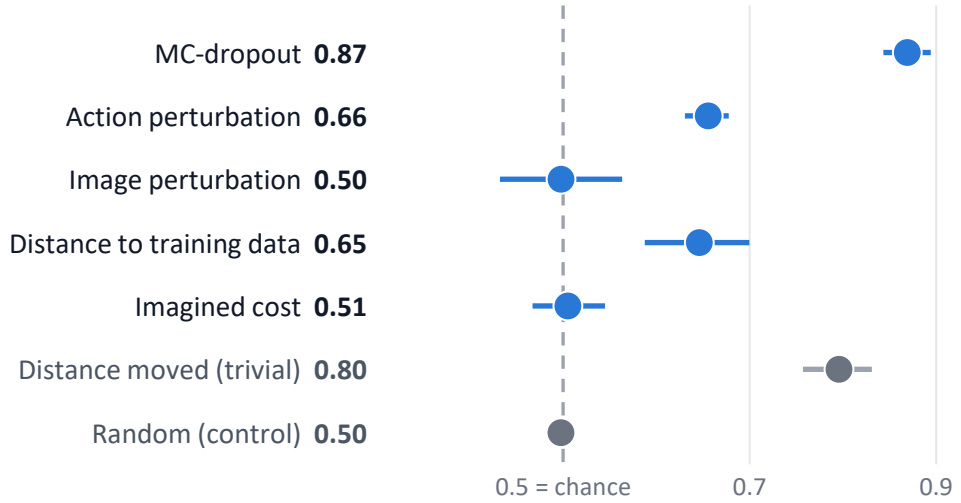
MEASURED PUBLISHED

Evidence so far: baseline reproduced, and a first trust-score pilot



H1 pilot: does a trust score flag the imagined steps with the largest real error?

AUROC with 95% interval · 40 Push-T decision points · 38,400 imagined steps checked against the simulator



3.8×

faster per 300-candidate rollout (26.2 s → 6.9 s); CEM's top 30 and its update unchanged

65% → 65%

pilot, offline: chosen plan succeeds, plain → MC-dropout gate; even the oracle gate gives 65%, so only the closed loop (H4) can show a gain

0.94

Wall success after round 3, mean of 2 seeds (published 0.96)

1.00

PointMaze success after round 2, 1 seed (published 0.98)

TARGET

Progress and next steps: baseline done, first trust-score pilot in

NOVEL COMPONENTS

- **C1** Error-labelled dataset of imagined vs real latents — PILOT P1
- **C2** Calibration test of ≥ 3 trust scores, e.g. ensembles [22], MC-dropout [23] — PILOT P2
- **C3** Gated CEM-MPC with the fallback ladder — PLANNED P3
- **C4** Backbone-agnostic interfaces; 8 GB reproduction recipe — BUILT

PROGRESS: 10 REVIEW WORK PACKAGES

4 done: requirements, literature review, environment and data pipeline, MDP formulation

4 partial: implementation (baseline runs and the pilot error logger; gate not built), evaluation (Push-T 3 seeds, Wall 2, PointMaze 1, H1 pilot), visualisation, documentation

2 not started: scorer training, final validation. No percentage claimed: the remaining packages hold most of the research effort.

ROADMAP (PROPOSED, RELATIVE WEEKS)



PILOT RESULT → NEXT

Best untrained score: MC-dropout, AUROC 0.87 [0.84–0.89] vs random 0.50: it beats chance and the trivial baseline (1 seed, 40 episodes). Next: 3 seeds, more sampling contexts, then H4 in the closed loop.

GO / NO-GO (END OF W6)

Build the closed loop only if a score beats the random and distance-to-start baselines on H1 (non-overlapping 95% CIs); otherwise report the negative result with the oracle-gate ceiling.

DONE WHEN

H1–H3 reported for ≥ 3 scores with bootstrap CIs; H4 on the same 50 pairs × 3 seeds as the baseline; every number regenerated from logs by script.

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[25] Fu et al. 2020 · D4RL (PointMaze) · preprint arxiv.org/abs/2004.07219

[26] DINO-WM code (MIT) @ 0a9492f and OSF data github.com/gaoyuezhou/dino_wm

Full entries with authors, venues and verified links: references.md and the companion doc. The 37-slide extended deck holds the detailed comparison, the three further architecture views and the appendix.